

Neurobiology of the Habenula

Functional Role of the Lateral Habenula

The lateral habenula (LHb) acts as a key node in the anti-reward system, firing when expected rewards are omitted. It projects signals to the rostromedial tegmental nucleus (RMTg) using glutamate as its primary neurotransmitter. In patients with severe refractory depression, deep brain stimulation (DBS) targeting the lateral habenula has been shown to reduce depressive symptoms by inhibiting LHb hyperactivity.